

**THE UNITED REPUBLIC OF TANZANIA
PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT**



**TABORA, SINGIDA, IRINGA AND DODOMA (TASIDO)
FORM TWO ZONAL ASSESSMENT**

031

Duration: 2:30 Hours

Year. 2026

INSTRUCTIONS

1. This paper consists of sections **A**, **B** and **C** with a total of **ten (10)** questions.
2. Answer **all** questions in the spaces provided.
3. Section **A** carries **fifteen (15)** marks, section **B** **seventy (70)** marks and section **C** carries **fifteen (15)** marks.
4. All writing must be in blue or black ink **except** drawings which must be in pencil.
5. All communication devices, calculators and any unauthorized materials are **not** allowed in the Mock Assessment room.
6. Write your **Mock Assessment Number** at the top right corner of every page.
7. Where necessary the following constants may be used:
 - (i) The earth acceleration due to gravity, $g = 10m/s^2$
 - (ii) Density of water = $1g/cm^3$ or $1000kg/m^3$

FOR ASSESSORS' USE ONLY		
QUESTION NUMBER	SCORE	ASSESSOR'S INITIALS
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTAL		
CHECKER'S INITIALS		

This paper consists of 8 printed pages

SECTION A: (15 MARKS)

Answer all questions in this section.

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the space provided.
 - (i) A specialized doctor at KCMC hospital wanted to measure the diameter of a needle of a syringe to an accuracy of three decimal places of centimetres. Which of the following instruments would you advise him to use?
 - A. Vernier calliper
 - B. Engineer's calliper
 - C. Doctor gauge tape measure
 - D. Micrometer screw gauge
 - (ii) Which of the following best describes why the knowledge of physics is the father of all-natural sciences?
 - A. Explains how energy passes from one object to another.
 - B. Explains the fundamental aspects of the universe.
 - C. Explains the motion of the objects that can be seen by naked eyes.
 - D. Explains how gravity works on earth.
 - (iii) The force of gravity is.
 - A. Contact force
 - B. Consequential force
 - C. Action-at-a-distance force
 - D. Reaction-at-a-normal force
 - (iv) Form one students were performing a smoke cell experiment in the laboratory. This experiment demonstrates which property of matter?
 - A. Osmosis
 - B. Brownian motion
 - C. Diffusion
 - D. Capillarity
 - (v) How does depth affect the pressure in liquids?
 - A. Pressure is inversely proportional to depth.
 - B. Pressure is not equal to depth.
 - C. Pressure is equal to depth.
 - D. Pressure is directly proportional to depth.
 - (vi) What happens when parallel rays of light fall on an irregular reflecting surface?
 - A. Reflected rays are not parallel.
 - B. Reflected rays are also parallel.
 - C. Reflected rays are parallel to the incident rays.
 - D. Angle of incidence is not equal to the angle of reflection.
 - (vii) Which instrument is used to detect the presence of electric charge on materials?
 - A. Electroscope
 - B. Gold-leaf
 - C. Electrophorus
 - D. Capacitor
 - (viii) Which of the following is the SI-unit of capacitance?
 - A. Ampere
 - B. Microfarad
 - C. Volt
 - D. Coulomb
 - (ix) The metallic rod, typically made of copper, which is used to protect buildings from lightning is called:
 - A. Thunder
 - B. Zigzag path
 - C. Lightning conductor
 - D. Cloud

(x) A capacitance of $1\ \mu\text{F}$, $4\ \mu\text{F}$, and $2\ \mu\text{F}$ are connected in series. Calculate the capacitance that could replace them.

A. $\frac{3}{2}\ \mu\text{F}$

C. $\frac{2}{7}\ \mu\text{F}$

B. $\frac{1}{2}\ \mu\text{F}$

D. $\frac{2}{9}\ \mu\text{F}$

2. Match the items in **LIST A** with those in **LIST B**.

LIST A	LIST B
(i) Ammeter	A. Measures pd
(ii) $I \propto l$	B. Measures current
(iii) Rheostat	C. Ohm
(iv) Series connection	D. Controls current
(v) Charge	E. Ohm's law
	F. Controls pd
	G. Coulomb
	H. Constant current
	I. Galvanometer
	J. Constant pd

Answers

LIST A	i	ii	iii	iv	v
LIST B					

SECTION B: (70 MARKS)

Answer **all** questions in this section.

3. (a) State the Archimedes principle.

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(b) An object of weight $18\ \text{N}$ immersed in water. It experiences an upthrust of $2\ \text{N}$. Calculate:

(i) Real weight.

(ii) Relative density.

(iii) Density of the object in kg.

4. (a) Distinguish between the density and the relative density of a substance. State the units in which density and relative density can be used.

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- (b) A certain school was performing an experiment of weighing sand. The following results were obtained:

Mass of empty density bottle	= 12.0 g
Mass of bottle + sand	= 62.0 g
Mass of bottle + sand + water	= 132.0 g
Mass of bottle filled with water	= 87 g

Calculate:

- (i) Relative density of sand.

- (ii) Density of sand.

- (iii) Volume of sand.

- (iv) Volume of bottle.

(a) Why it is easier to cut a bar of soap using a thin piece of wire than a thick one?

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(b) A car of mass 8 000 kg has one of its tyres with an area of 50 cm² in contact with the ground. If this is the four-wheel drive vehicle, calculate the pressure exerted on the ground by the car.

6.

(a) (i) Give two examples of kinetic energy in real life situations.

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(ii) State two factors which determine kinetic energy of a body.

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(b) Anna has a model car of mass 20 kg moves at a speed of 6 m/s for 10 s. Find the power developed by its engine.

(i) In Watts;

(ii) In Horsepower.

(a) When water and Mercury were in two separate measuring cylinders, the teacher asked students to observe the reading in both cylinders and found that meniscus of water bent downward and that of mercury bent upward. Why?

(i) Meniscus of water bend downward?

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(ii) Meniscus of mercury, bend upward?

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(b) A force of 4 N causes a certain copper wire to extend to 1.0 mm. Find the force that will cause 3.2 mm extension on the same wire.

8.

(a) Give two differences between speed and velocity as applied in linear motion.

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(b) A car starts from rest and accelerates uniformly to 20 m/s for 5 seconds. It maintains a constant velocity for 20 seconds. The brakes are then applied and the car decelerates to rest for 3 seconds.

(i) Sketch the velocity–time graph.

(ii) Find the total distance covered.

9. (a) State four ways in which magnets are used in daily life.

- (i)
- (ii)
- (iii)
- (iv)

(b) In which part of the fridge or microwave do we have magnetic strips installed? Why?

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(c) Mention any four properties of magnet.

- (i)
- (ii)
- (iii)
- (iv)

(d) How can you test for the polarity a given magnet. Explain why repulsion is sure test?

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SECTION C: (15 MARKS)

Answer this question.

10. (a) Sketch a graph showing:

- (i) Charging a capacitor.

(a) Discharging a capacitor.

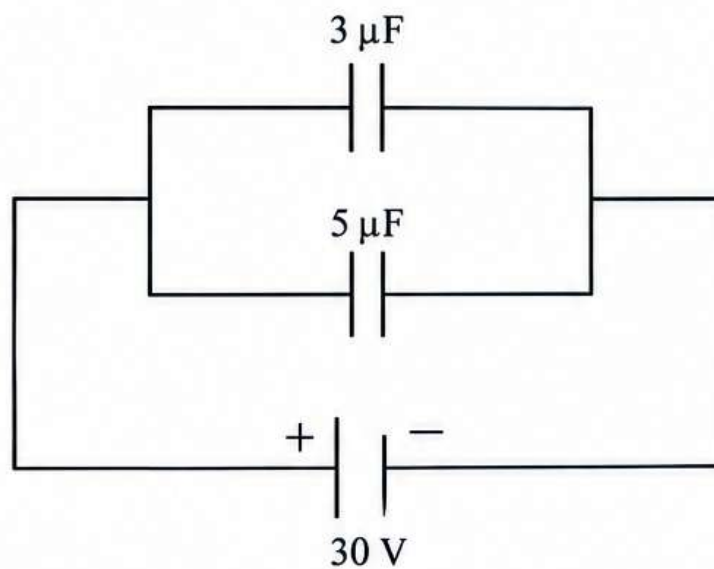
(b) The capacitance of a parallel plate capacitor is affected by three major factors, name them.

(i)

(ii)

(iii)

(c) Study the circuit below and then answer the questions that follow.



(i) Calculate the total capacitance.

(ii) How much charge is stored?